P11 Shooting

CRE132 worksheet

What you'll make

A platformer hero who throws fireballs. Hold **J** (or the left mouse button, or the gamepad's East button) to throw them the way you face, four a second. Cats take 3 hits: each hit makes them flash and stop for a moment, and the third turns them into a floating ghost.

Most of this is already built from earlier samples: S07's object pools and shooter, S01's Health, S09's events and P08's patrol. You'll add one small script that makes hits *feel* like they land, and see how separate systems combine with almost no new code.

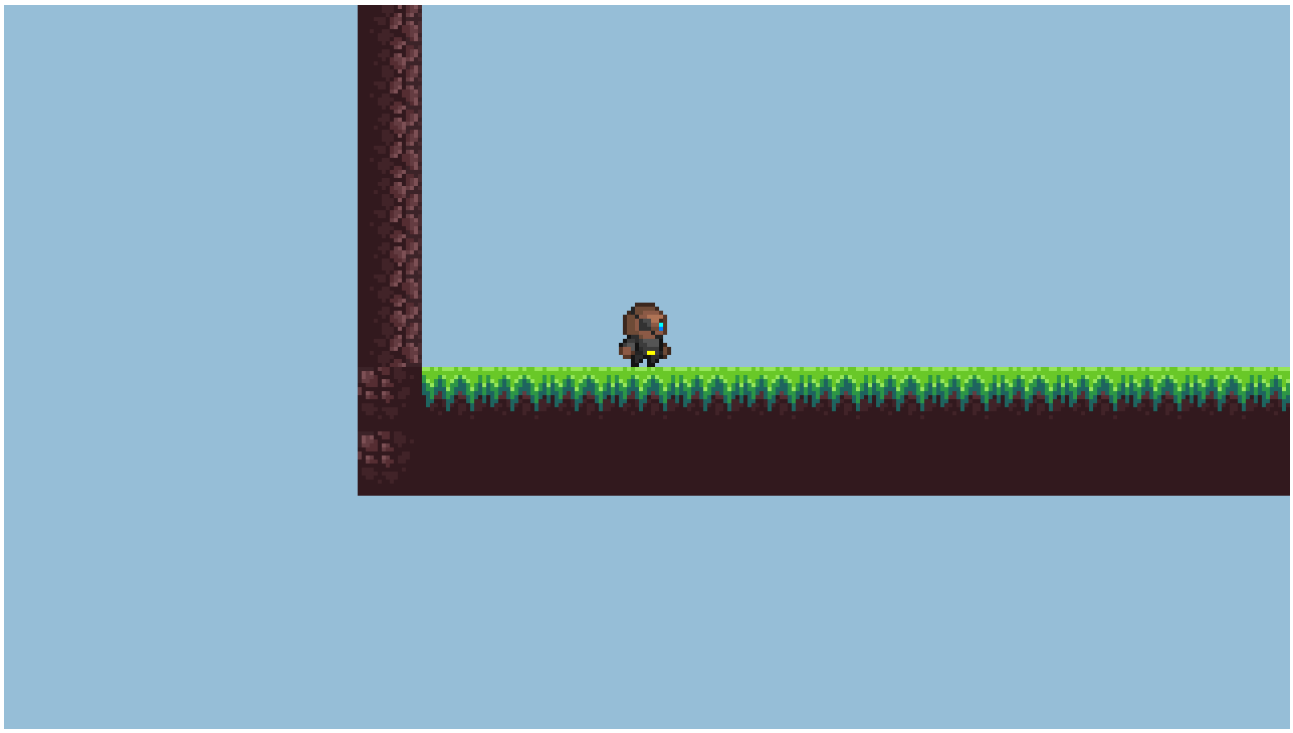


Figure 1: The start of the level: the player by the left wall, ready to throw fireballs.

Before you start

1. Unzip `CRE132_P11_Shooting_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_P11_Shooting_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Platformer/P11_Shooting`.
3. Press **Play**. Run (A/D or arrows), jump (Space) and hold **J** to throw fireballs. Three hits and a cat just vanishes, with no reaction to the first two. Touching a cat still hurts you, with P09's knockback. Press Play again to stop.

Look at what's already there:

- The **Player** has S07's **Player Shooter**, with **Fireball Pool** and **Impact Pool** in its slots. It reads the sprite's `flipX` (which P06's animator sets) to know which way to shoot. **Time Between Shots** is **0.25**, four a second; S07 used 0.12 for a stream.
- Each of the seven **Cat** objects (in **Enemies**) has **Health** with **Max Health** 3. S07's `Projectile` already damages anything with Health.

Making a script: in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. The hit reaction: EnemyHitReaction

Make a folder `Assets/_game/Scripts/P11_Shooting` and create `EnemyHitReaction` in it:

```
// How an enemy reacts when hit: a quick see-through flash and a short stagger. When it
// dies, a ghost floats away. (Tinting a sprite can only darken it; a true white flash
// needs a shader, see E04/E03.) Health's On Damaged event calls Flinch(); On Died calls
// SpawnGhost().
using System.Collections;
using UnityEngine;

public class EnemyHitReaction : MonoBehaviour
{
    [SerializeField] private Color flashColour = new Color(1f, 1f, 1f, 0.35f);
    [SerializeField] private float flashTime = 0.1f;
    [SerializeField] private float staggerTime = 0.2f;
    [SerializeField] private GameObject ghostPrefab;
    [SerializeField] private float ghostTime = 0.6f;

    private SpriteRenderer spriteRenderer;
    private Rigidbody2D body;
    private EnemyPatrol patrol;

    private void Awake()
    {
        spriteRenderer = GetComponent<SpriteRenderer>();
        body = GetComponent<Rigidbody2D>();
        patrol = GetComponent<EnemyPatrol>();
    }

    public void Flinch()
    {
        StartCoroutine(FlinchRoutine());
    }

    private IEnumerator FlinchRoutine()
    {
        // Stop walking for a moment and flash.
        patrol.enabled = false;
        body.linearVelocity = Vector2.zero;
        spriteRenderer.color = flashColour;

        yield return new WaitForSeconds(flashTime);
        spriteRenderer.color = Color.white;

        yield return new WaitForSeconds(staggerTime - flashTime);
        patrol.enabled = true;
    }

    public void SpawnGhost()
    {
        GameObject ghost = Instantiate(ghostPrefab, transform.position,
            Quaternion.identity);
        ghost.GetComponent<SpriteRenderer>().flipX = spriteRenderer.flipX;
        Destroy(ghost, ghostTime);
```

```
    }
}
```

What's going on:

- **`Flinch` and `SpawnGhost` are public** because Health's events will call them. Neither method knows anything about fireballs: they run whenever the cat is hurt or dies, whatever did it.
- **`FlinchRoutine`** switches the patrol off and stops the body, so the cat stops walking. It turns the sprite see-through for `flashTime` (0.1 s), then waits the rest of `staggerTime` (0.2 s in total) before letting it walk again.
- **A see-through flash** is used because tinting a sprite can only make it darker. A true white flash needs a shader (E04, E03).
- **`SpawnGhost`** is P08's ghost: spawned where the cat was, facing the same way, and removed after `ghostTime`.

Add it: select all seven **Cat** objects (click the first in **Enemies**, Shift-click the last) and **Add Component > Enemy Hit Reaction**. Leave **Flash Colour** (white with alpha 0.35), **Flash Time** 0.1, **Stagger Time** 0.2 and **Ghost Time** 0.6. Drag the **Cat Ghost** prefab from `Assets/_game/Prefabs/Characters` into the **Ghost Prefab** slot.

2. Connecting it with Health's events

Health calls its **On Damaged** event on every hit and **On Died** when health reaches zero (S09). Connect them to the new methods.

Select the first **Cat** and find its **Health** component. Unity can't edit events on several objects at once, so do this for each cat in turn:

1. **On Damaged ()** is empty: click **+** to add a row. Drag this **Cat** from the Hierarchy into the row's object slot, then choose **EnemyHitReaction > Flinch ()** from the function list.
2. **On Died ()**: the same, but choose **EnemyHitReaction > SpawnGhost ()**.

Make sure each cat's slots point at that same cat, not the first one you did.

Test it: press Play and shoot a cat. It flashes and stops for a moment with each hit, and the third hit turns it into a ghost that floats away.

Check your work

- ☐ Holding J throws fireballs the way you face, four a second.
- ☐ Each hit makes a cat flash see-through and stop briefly, then it walks on.
- ☐ The third hit turns the cat into a floating ghost.
- ☐ Touching a cat still hurts you and knocks you back.
- ☐ Compare with `CRE132_P11_Shooting_Finished` if something behaves differently.

Key points

- **Systems built separately combine with almost no new code.** Pooling, health, events and patrol were each written for one job, and they work together here.
- **A flinch makes hits feel like they land.** Without it, enemies seem to ignore your shots.
- **Events keep scripts apart.** Health doesn't know `EnemyHitReaction` exists; the connection lives in the Inspector.

If something goes wrong

- **The cats don't react:** the **On Damaged** row on that cat is empty or set to **No Function**.

- **Shooting one cat makes a different cat flinch:** its event slots point at the wrong cat. Each cat's events must use that cat.
- **NullReferenceException in SpawnGhost:** the **Ghost Prefab** slot is empty.
- **A cat stops for ever after a hit:** check `patrol.enabled = true;` is at the end of `FlinchRoutine`, and that **Stagger Time** is bigger than **Flash Time**.
- **No fireballs:** the **Player Shooter**'s pool slots or **Attack Action** slot have been cleared.

Try it

1. Give some cats **Max Health** 1 and some 5.
2. Knock the cat back a little in the direction the fireball was going.
3. Shoot upwards when Up is held with Attack.